Oral Feeding Requires Suck–Swallow–Breathe Coordination

Safe oral feeding is a maturational coordination of sucking, swallowing, and respiration; the elements mature at different rates, especially in preterm infants, and feeding method is not parental worth.

ADE: HIGH-COORDINATION; MODERATE-PRACTICE-EFFECTS | STATUS: DRAFT | DOMAIN: BIRTH, FEEDING | REPLICATION: VERIFI

WHAT WE KNOW (Narrowest Supported Conclusion)

After birth, nutrition must move from placental transfer to oral feeding. That requires sucking, swallowing, breathing, arousal, and gastrointestinal function to work together. Each function has parts that mature at different times. “Coordination” is therefore not a single switch. Preterm infants often take longer to feed safely by mouth; swallow–respiration matching is a common bottleneck.

EVIDENCE AND CAUSAL ASSESSMENT

Lau (2015), Annals of Nutrition and Metabolism (PMC4530609; author’s neonatology group), reviews nutritive sucking (suction and expression), swallow processing, and respiration, arguing that readiness is not well defined clinically and that nonnutritive sucking is not sufficient evidence of oral-feeding readiness. Designs are physiologic observation and preterm/term comparison, not randomized feeding-method trials for later IQ. breastfeeding-cognitive-development.md remains the place for PROBIT/sibling cognition.

EFFECT SIZE AND PERSISTENCE

The immediate outcome is safe transfer of a bolus. Persistence of feeding difficulty is clinical and heterogeneous. There is no license to read feeding method as a character or intelligence result.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (maturational oral-motor physiology, not an observational population phenotype association).

Gestational age, illness, congenital difference, posture, and caregiver/clinical support are intertwined. Breast, bottle, expressed milk, and tube feeding are distinct situations.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Feeding difficulty is not a verdict on parental worth. Breastfeeding associations with later cognition are a separate, heavily confounded literature.

WHAT ACTION IS REASONABLY JUSTIFIED

Skilled feeding support where wanted or needed, and evaluation of unsafe feeding. Ranking families by feeding method is not justified here.

VERIFIED SOURCES & DOI IDENTIFIERS

- Lau (2015), Annals of Nutrition and Metabolism (PMC4530609)